Question 1 : What is Simple Linear Regression (SLR)? Explain its purpose.
Ans:- **Simple Linear Regression (SLR)** is a statistical and machine learning technique used to model the relationship between **one independent variable (X)** and **one dependent variable (Y)**. It predicts the value of the dependent variable based on the independent variable by fitting the best straight line through the data.

The mathematical equation of SLR is:

$$Y=a+bX$$

Where:

- **Y** = Dependent (target) variable
- **X** = Independent (predictor) variable
- **a** = Intercept (value of Y when X = 0)
- **b** = Slope (change in Y for every one-unit increase in X)

Purpose of Simple Linear Regression

The main purposes of Simple Linear Regression are:

1. **Predict values** of a dependent variable using one independent variable.
2. **Understand the relationship** between two variables.
3. **Measure the effect** of the independent variable on the dependent variable.
4. **Support decision-making** by identifying trends and making forecasts.

Example

Suppose a company wants to predict **house prices** based on the **area of the house**.

- **Independent Variable (X):** House Area (square feet)
- **Dependent Variable (Y):** House Price

SLR helps estimate the house price for any given area by fitting a straight-line equation.

Applications

- Predicting house prices
- Estimating sales based on advertising expenditure
- Forecasting employee salaries based on years of experience
- Predicting exam scores based on study hours

Question 2: What are the key assumptions of Simple Linear Regression?
Ans:- Simple Linear Regression (SLR) is based on several assumptions. These assumptions ensure that the model produces accurate and reliable results.

1. **Linearity**
    - There should be a linear relationship between the independent variable (X) and the dependent variable (Y).
    - This means that changes in X should result in proportional changes in Y.
2. **Independence of Errors**
    - The observations should be independent of each other.
    - The error (residual) for one observation should not influence the error of another.
3. **Homoscedasticity (Constant Variance)**
    - The residuals should have constant variance across all levels of the independent variable.
    - If the variance changes, it is called **heteroscedasticity**, which can reduce the model's reliability.
4. **Normality of Residuals**
    - The residuals (errors) should be approximately normally distributed.
    - This assumption is important for hypothesis testing and confidence intervals.
5. **No Significant Outliers**

- The dataset should not contain extreme outliers that disproportionately influence the regression line.
- Outliers can distort the model and reduce prediction accuracy.

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Ans:- The mathematical equation of a **Simple Linear Regression (SLR)** model is:

\hat{y} = b_0 + b_1x

^=b0+b1X

Where:

- **Y^\hat{Y}Y^** = Predicted value of the dependent variable (target variable).
- **XXX** = Independent variable (predictor variable).
- **b0b_0b0** = Intercept (the predicted value of YYY when X=0X = 0X=0).
- **b1b_1b1** = Slope or regression coefficient (the change in the predicted value of YYY for every one-unit increase in XXX).

Question 4: Provide a real-world example where simple linear regression can be applied.

Ans:- A common real-world application of **Simple Linear Regression (SLR)** is **predicting house prices based on the size (area) of the house**.

- **Independent Variable (X):** House Area (in square feet)
- **Dependent Variable (Y):** House Price

By analyzing historical data, a regression model can learn the relationship between house area and price. Once the model is trained, it can estimate the price of a new house based on its area.

For example, if the regression equation is:

Y^=50,000+200X

Where:

- Y^\hat{Y}Y^ = Predicted house price (₹)
- XXX = House area (square feet)

If a house has an area of **1,500 square feet**, the predicted price is:

Y^=50,000+200(1,500)=350,000

So, the estimated house price is **₹350,000**.

Other Real-World Applications

- Predicting sales based on advertising expenditure.
- Estimating employee salary based on years of experience.
- Predicting exam scores based on study hours.
- Forecasting electricity consumption based on temperature.

Question 5: What is the method of least squares in linear regression?

Ans:- The **Method of Least Squares** is a mathematical technique used in **Simple Linear Regression** to determine the **best-fit line** for a set of data points. It finds the regression line by minimizing the **sum of the squared differences (residuals)** between the actual values and the predicted values.

Explanation

- **Actual Value (Y):** The observed value from the dataset.
- **Predicted Value (Y^):** The value estimated by the regression model.
- **Residual (Error):** The difference between the actual and predicted values.

Residual=Y−Y^

The method minimizes the **Sum of Squared Errors (SSE**

SSE=i=1∑n(Yi−Y^i)

Squaring the errors ensures that:

- Positive and negative errors do not cancel each other.
- Larger errors receive greater penalties.
- The best-fitting regression line is obtained.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Ans:- **Logistic Regression** is a supervised machine learning algorithm used for **classification problems**. It predicts the **probability** that an observation belongs to a particular class, such as **Yes/No**, **True/False**, or **0/1**. It uses the **Sigmoid (Logistic) Function** to convert the output into a probability between **0 and 1**.

The logistic regression equation is:

P(Y=1)=1+e−(b0+b1X)

Where:

- **P(Y=1)P(Y=1)P(Y=1)** = Probability that the outcome belongs to class 1.
- **b0**= Intercept.
- **b1** = Regression coefficient (slope).
- **X**= Independent variable.
- **e** = Euler's number (approximately 2.718).

Purpose of Logistic Regression

- Predicts binary outcomes (e.g., Pass/Fail, Spam/Not Spam).

- Estimates the probability of an event occurring.
- Classifies observations into different categories.

Linear Regression

- Predicting house prices based on area.
- Predicting sales based on advertising expenditure.
- Estimating salary based on years of experience.

Logistic Regression

- Predicting whether an email is spam or not.
- Predicting whether a customer will purchase a product (Yes/No).
- Predicting whether a patient has a disease (Positive/Negative).
- Predicting whether a student will pass or fail an exam.

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Ans:- Regression models are evaluated using metrics that measure how closely the predicted values match the actual values. Three of the most common evaluation metrics are:

1. Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average of the absolute differences between the actual and predicted values. It treats all errors equally and is easy to interpret.

Formula:

MAE=1n∑i=1n∣Yi−Y^i

Interpretation:

- Lower MAE indicates better model performance.
- A value of **0** means perfect predictions.

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2. Mean Squared Error (MSE)

Mean Squared Error (MSE) calculates the average of the squared differences between actual and predicted values. Squaring the errors gives more weight to larger errors.

Formula:

MSE=1n∑i=1n(Yi−Y^i)2

Interpretation:

- Lower MSE indicates a better-fitting model.
- Large prediction errors are penalized more heavily than small ones.

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3. Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is the square root of the Mean Squared Error. It measures the average prediction error in the same units as the target variable.

Formula:

RMSE=1n∑i=1n(Yi−Y^i)2

Interpretation:

- Lower RMSE indicates better prediction accuracy.
- RMSE is easier to understand because it is expressed in the same units as the dependent variable.

Question 8: What is the purpose of the R-squared metric in regression analysis?

Ans:- **R-squared (R^2)**, also known as the **Coefficient of Determination**, is a statistical metric used to evaluate how well a regression model fits the data. It measures the proportion of the variation in the dependent variable that is explained by the independent variable(s).

$R2=1-\sum(Yi-\bar{Y})2/\sum(Yi-\hat{Y}i)2$

Where:

- **Yi**= Actual value
- **$\hat{Y}i$** = Predicted value
- **$\bar{Y}$** = Mean of the actual values
- **$\sum(Yi-\hat{Y}i)$** = Sum of Squared Errors (SSE)
- **$\sum(Yi-\bar{Y})2$** = Total Sum of Squares (SST)

Purpose of R-squared

- Measures how well the regression model explains the variation in the target variable.
- Evaluates the goodness of fit of the regression model.
- Helps compare different regression models.
- Indicates how much of the variability in the dependent variable is explained by the independent variable(s).

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept. (Include your Python code and output in the code box below.)

Ans:-

```
# Import required libraries

import numpy as np

from sklearn.linear_model import LinearRegression

# Sample dataset

X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)   # Independent variable

y = np.array([2, 4, 5, 4, 5])                 # Dependent variable
```

```
# Create and train the model

model = LinearRegression()

model.fit(X, y)

# Print the slope and intercept

print("Slope (Coefficient):", model.coef_[0])

print("Intercept:", model.intercept_)
```

Output : Slope (Coefficient): 0.6

Intercept: 2.2

Question 10: How do you interpret the coefficients in a simple linear regression model?

Ans:- In a **Simple Linear Regression (SLR)** model, the coefficients describe the relationship between the independent variable (**X**) and the dependent variable (**Y**).

The regression equation is:

Y^=b0+b1X

Where:

- **Y^**= Predicted value of the dependent variable
- **b0**= Intercept
- **b1** = Slope (Regression Coefficient)
- **X** = Independent variable

. Intercept (b_0)

Interpretation:

The intercept represents the predicted value of Y when X = 0.

Example:

If the model is:

$\hat{y} = 50 + 8X$

Then:

- $b_0 = 50$
- When X = 0, the predicted value of Y is 50.

For example:

If X represents years of experience and Y represents salary, the model predicts a salary of 50 units for someone with 0 years of experience.

2. Slope (b_1)

Interpretation:

The slope indicates the average change in Y for every one-unit increase in X.

Example:

$\hat{y} = 50 + 8X$

Here:

- $b_1 = 8$
- For every 1-unit increase in X, Y increases by 8 units on average.